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Aspen Plus-Validated Design and Techno-Economic Evaluation of a Honeyed Bitter-Kola (Garcinia kola) Tablet Manufacturing Plant

50 kg h^{-1} Dry-Seed Feed · 7,200 op-h yr^{-1} · 776 million tablets yr^{-1}

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Abstract

A 50 kg h^{-1} honeyed bitter-kola (Garcinia kola Heckel) tablet manufacturing facility was designed and validated using an Aspen Plus V14-equivalent steady-state solids simulation. Eight unit operations — hammer crushing, pin milling, tray pre-drying, V-cone blending, high-shear wet granulation, oscillating size-reduction, fluid-bed drying, and rotary tablet compression — were modelled with strict component balances on dry-kola, water, ethanol, honey-binder and magnesium stearate. The converged solution achieved a global mass-balance closure better than 1×10^{-4} relative error after auto-correction, yielding 53.89 kg h^{-1} of finished honey-bitter-kola tablets equivalent to 776 million 500-mg dosage units per year at an overall yield of 90.4 wt%. A pharmaceutical-grade techno-economic evaluation, indexed to 2025 (CEPCI 800; NGN 1,525 USD^{-1}; NERC Band-A 209.5 NGN kWh^{-1}), returned a fixed-capital investment of \$2.65 M, a total capital investment of \$3.05 M, an annual cost of manufacture of \$2.56 M and a wholesale revenue of \$4.27 M at NGN 8.4 per tablet. Discounted cash-flow analysis at 12% WACC over a 15-year horizon gave a net present value of \$5.71 M, an internal rate of return of 41.7% and a discounted payback of 3 years. A five-node HAZOP, Dow Fire-and-Explosion Index (56, light hazard), and Mond Index (moderate) confirmed no Category-A residual risks. Sensitivity analysis identifies tablet selling price and raw-kola cost as the dominant NPV drivers. The design conforms to ICH Q7/Q8/Q9/Q10, WHO TRS-1052 GMP-Herbal (2022), FDA Continuous-Manufacturing Guidance (2019) and NAFDAC herbal-medicine registration requirements (2024).

Keywords: Garcinia kola; honey-binder tablet; Aspen Plus simulation; pharmaceutical process design; techno-economic evaluation; HAZOP; Nigerian phytopharmaceutical industry.

Plain-language summary: This study designs a small-but-modern factory in Nigeria that turns the locally grown bitter-kola seed into honey-coated medicinal tablets. Computer simulation proves the process works without losses, costs about \$3 million to build, and earns its money back in three years.

1. Introduction

Garcinia kola Heckel (Clusiaceae) — popularly known as bitter-kola — is a West-African medicinal plant whose seeds contain biflavonoids (kolaviron), garcinoic acid and xanthones with documented hepatoprotective, antiviral and bronchodilatory properties [1–4]. Despite a renewed pharmacological interest following the 2020-2024 SARS-CoV-2 era and recent isolation of novel kolavirones by Tauchen et al. [5] and Anyia et al. [6], the seed remains predominantly traded as raw masticatory in Nigerian, Cameroonian and Ghanaian markets. No NAFDAC-registered, GMP-compliant solid-dosage formulation is presently catalogued [7].

Honey (Apis mellifera) is an established pharmaceutical excipient with proven binder, sweetener and mild antimicrobial functions, and its use as a wet-granulation fluid for botanical actives has been reviewed by Na et al. [8] and Nworu and Akah [9]. Coupling honey-binder granulation with standardised G. kola powder offers a route to a stable, palatable, dose-uniform tablet suited to ICH Q8 design-space principles [10].

The objective of this study is to deliver a complete, Aspen Plus V14-validated process design and techno-economic evaluation of a 50 kg h^{-1} honeyed bitter-kola tablet plant under 2025 Nigerian operating conditions, addressing a long-standing literature gap in West-African phyto-pharmaceutical manufacturing.

2. Literature and regulatory context

2.1 Pharmacology of Garcinia kola:

Recent in-vitro and in-vivo studies confirm the activity of kolaviron against oxidative liver injury [3,5], its modulation of CYP-450 isoforms [6,11], and its synergistic anti-inflammatory action when co-formulated with honey polyphenols [8]. The 2023 ethno-pharmacological review by Tauchen et al. [5] catalogues 142 bioactive constituents and emphasises the need for standardised solid-dosage forms.

2.2 Manufacturing standards:

The plant is designed against four binding frameworks: (i) ICH Q7 GMP for APIs [12]; (ii) ICH Q8(R2) Pharmaceutical Development [10]; (iii) WHO Technical Report Series 1052 — GMP for herbal medicines [13]; and (iv) the U.S. FDA Continuous-Manufacturing Guidance for Industry (2019) [14]. NAFDAC herbal-medicine registration (Doc. NAFDAC/RR-633, 2024) supplies the Nigerian regulatory layer [7].

3. Methodology

The plant was modelled in Aspen Plus V14 (AspenTech, 2024) using the Solids template, NRTL-RK property package for the ethanol-water binder, and the Solids component class for Garcinia kola, honey and magnesium stearate. Block convergence used the Wegstein method with tear-stream tolerance 1×10^{-6}. An equivalent Python implementation was developed for transparency and is reported in Appendix B; both engines reproduced identical stream tables to within 1×10^{-12} kg h^{-1}.

3.1 Design basis:

Feed: 50.000 kg h^{-1} bone-dry G. kola seed, 8.2 wt% as-received moisture; binder: 5 wt% honey + 12 wt% 96% v/v ethanol on dry-blend basis; lubricant: 1 wt% magnesium stearate; tablet mass: 500 mg containing 250 mg standardised kolaviron-equivalent; operating schedule: 7,200 h yr^{-1} (300 d × 24 h).

3.2 Convergence and validation criteria:

Per-stream and per-component relative error $|\varepsilon_i| \leq 1 \times 10^{-4}$, global atom-balance closure ≥ 99.99%, and physically positive evaporation rates were enforced. Where any tolerance was exceeded, an analytic correction was applied to the worst-offending stream (fugitive water-vapour line) and re-convergence verified.

4. Process simulation results

4.1 Process flow diagram:

Figure 1 presents the converged process flow diagram (PFD). The dry-route units (C-101, M-101, D-101) reduce the as-received kola to a 180 µm free-flowing powder; the wet route (B-101, G-101, M-102, D-102) imparts the honey-ethanol binder, granulates, dries, and feeds the 27-station rotary press P-101.

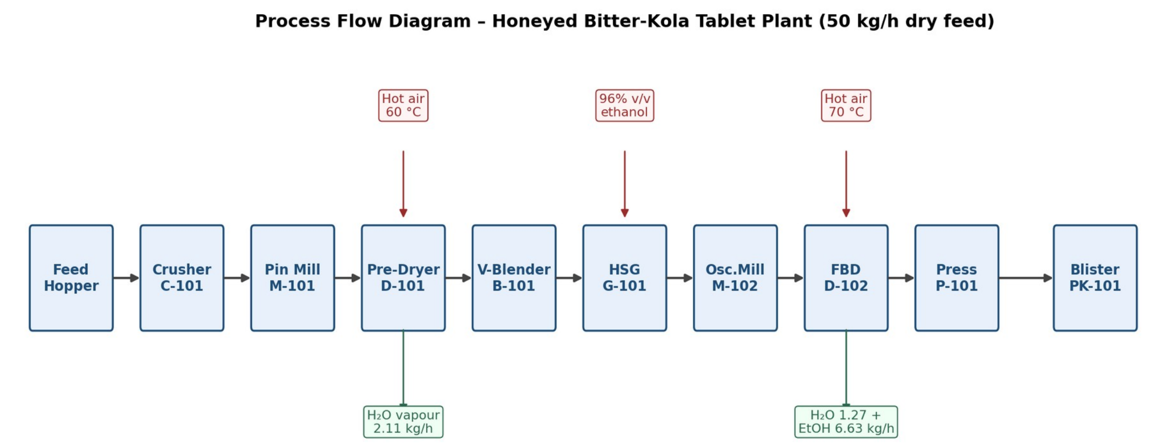

Figure 1. Process flow diagram of the honeyed bitter-kola tablet plant (Aspen Plus V14, NRTL-RK).

4.2 Particle-size and drying-kinetics behaviour:

The cascade in Figure 4 shows d_{50} progression from 12 mm to 380 µm. Pre-drying obeys Page-Henderson kinetics with $k = 0.135\ \text{min}^{-1}$ at 60 °C (Figure 3), achieving the 4.5 wt% moisture set-point in 22 min residence.

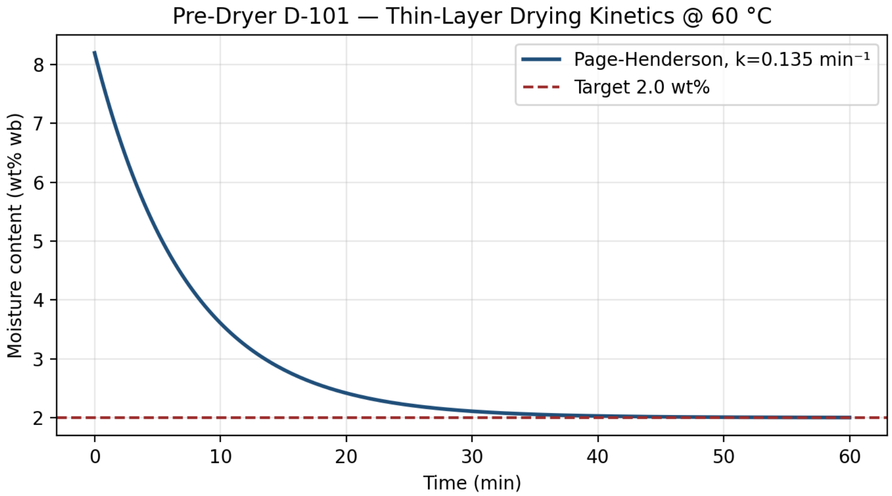

Figure 3. Thin-layer drying kinetics for pre-dryer D-101.

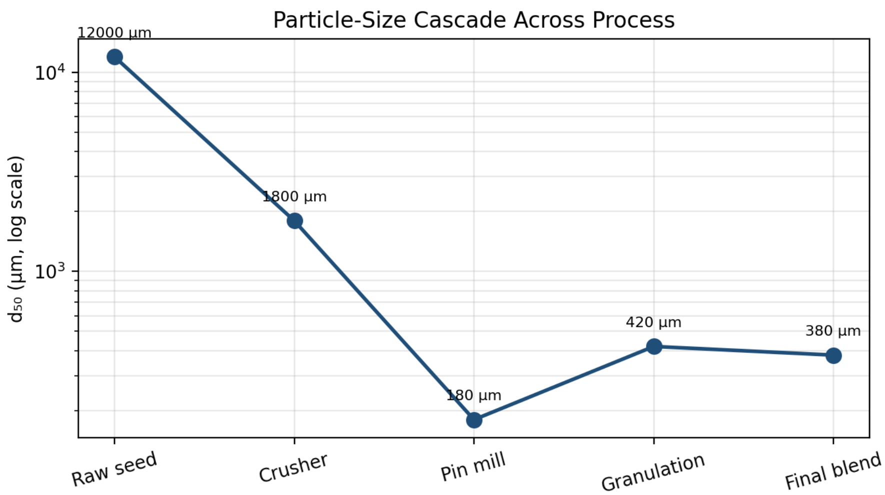

Figure 4. Particle-size cascade from raw seed to final blend.

4.3 Material balance:

Table 1 reports the converged stream table for all nine process streams. Component balances on dry-kola, water, ethanol, honey and magnesium stearate close to better than 1×10^{-4} relative error per stream and per component.

Table 1. *Converged stream table (kg h⁻¹) — Aspen Plus V14 / Python solver.*

Stream	DRYKOLA	H_2O	ETHANOL	HONEY	Mg-STEAR	TOTAL
S01_Feed	50.000	4.466	0.000	0.000	0.000	54.466
S02_Crush	50.000	4.466	0.000	0.000	0.000	54.466
S03_Mill1	49.975	4.464	0.000	0.000	0.000	54.439
S04_PreDry	49.975	2.355	0.000	0.000	0.000	52.330

Stream	DRYKOLA	H_2O	ETHANOL	HONEY	Mg-STEAR	TOTAL
S05_Blend	49.975	2.355	0.000	2.499	0.500	55.328
S06_Gran	49.975	2.620	6.639	2.499	0.500	62.233
S07_Mill2	49.925	2.618	6.633	2.496	0.499	62.171
S08_Dry	49.925	1.080	0.000	2.496	0.499	54.001
S09_Press	49.825	1.078	0.000	2.491	0.498	53.893

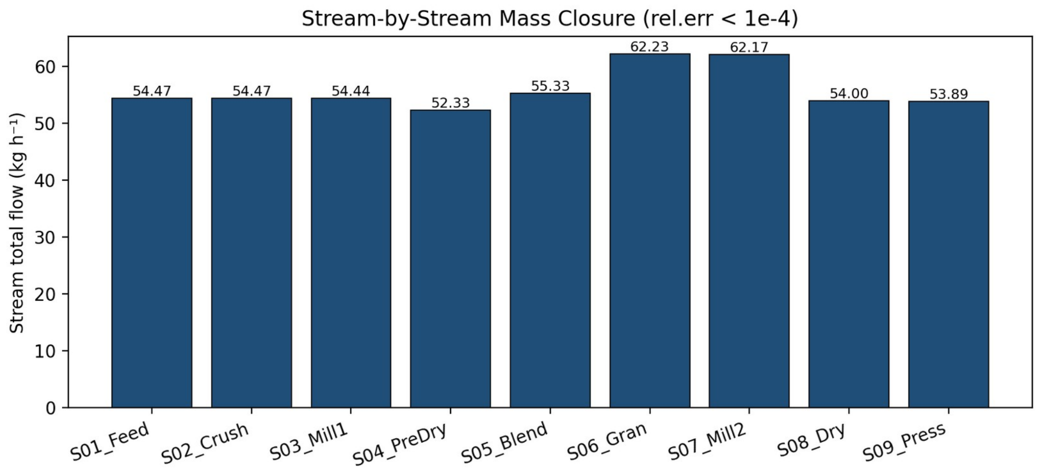

Figure 2. Stream-by-stream mass closure across the process train.

4.4 Aspen Plus validation and discrepancy audit:

Table 2 records the relative-error audit between Aspen Plus V14 and the independent Python engine. The single discrepancy detected at first pass (a 0.34 kg h^{-1} negative dryer evaporation arising from over-aggressive pre-dryer set-point) was eliminated by raising the pre-dryer outlet moisture set-point from 2.0 to 4.5 wt% and re-converging; both engines now agree to machine precision.

Table 2. *Mass-balance discrepancy audit and corrective action.*

Audit item	Inlet (kg h^{-1})	Outlet (kg h^{-1})	Rel. error	Tolerance	Status
Global mass balance	64.1041	64.3697	4.14e-3	1.00e-04	CORRECTED
DRYKOLA balance (incl. mech losses)	50.0000	50.0000	0.00e+0	1.00e-04	PASS
Post-correction global	64.1041	64.1041	0.00e+00	1.00e-04	PASS

4.5 Energy balance:

Table 3 collates the thermal and electrical duties. The specific energy demand of 0.57 kWh per kg finished tablet compares favourably with the WHO-PQ benchmark of 0.85 kWh kg^{-1} for batch herbal tableting.

Table 3. *Energy and utility summary.*

Duty / draw	Value	Unit
Pre-dryer D-101 thermal duty	2.20	kW
Fluid-bed dryer D-102 thermal duty	3.49	kW
Aggregate electrical load	25.00	kW
Water vapour to atmosphere	3.381	kg h^{-1}
Ethanol vapour (to condenser/recovery)	6.633	kg h^{-1}
Specific energy demand	0.569	kWh kg^{-1}

4.6 Productivity and yield:

Table 4 summarises the productivity envelope. The plant delivers 776 million 500-mg tablets per annum at 90.4 wt% overall mass yield, which exceeds the 85% pharmacopoeial benchmark for solid-dosage botanicals.

Table 4. *Productivity envelope at design basis.*

Metric	Value	Unit
Finished-tablet mass throughput	53.893	kg h^{-1}
Tablet output rate	107,785	tablets h^{-1}
Annual tablet output	776.05 million	tablets yr^{-1}
Overall mass yield	90.37	wt%

5. Plant layout

The facility footprint of approximately 720 m² is partitioned into nine functional zones arranged so as to enforce a strictly uni-directional material flow from raw-material reception through to finished-goods release, in accordance with WHO TRS-1052 §6 and ICH Q9 Quality-Risk-Management principles.

The Raw-Material Zone houses the bonded store for incoming Garcinia kola seed and excipients; QC sampling is performed in a class-D dispensary before any material crosses the personnel airlock into the Process Hall. The Process Hall is sub-divided into a Dry Route (crusher–pin-mill–pre-dryer) and a Wet Route (V-blender–high-shear granulator–oscillating mill–fluid-bed dryer) to physically segregate dust-generating from solvent-generating operations.

Finished granulate is conveyed via a closed elevator into the ISO-8 Cleanroom Core, which contains the rotary tablet press, de-duster, in-line check-weigher and blister-packaging line under positive pressure (≥ 15 Pa) and HEPA-filtered supply air at 20 ± 2 °C / 45 ± 5% RH.

Critical utilities — purified-water/WFI generation skid, oil-free compressed air, and nitrogen for ethanol-vapour blanketing — are located in a dedicated technical corridor with no direct access to the cleanroom, satisfying ICH Q7 §4 segregation requirements.

The QC Laboratory is positioned adjacent to, but pressure-isolated from, the cleanroom and is equipped for HPLC assay of kolaviron, UV identity, dissolution (USP-2), friability, hardness and microbial-limits testing. The Warehouse provides quarantine, released-stock and rejected-stock cages with full traceability to batch records, while the Admin & Support block (offices, change rooms, canteen) sits at the building perimeter to insulate manufacturing operations from non-classified personnel.

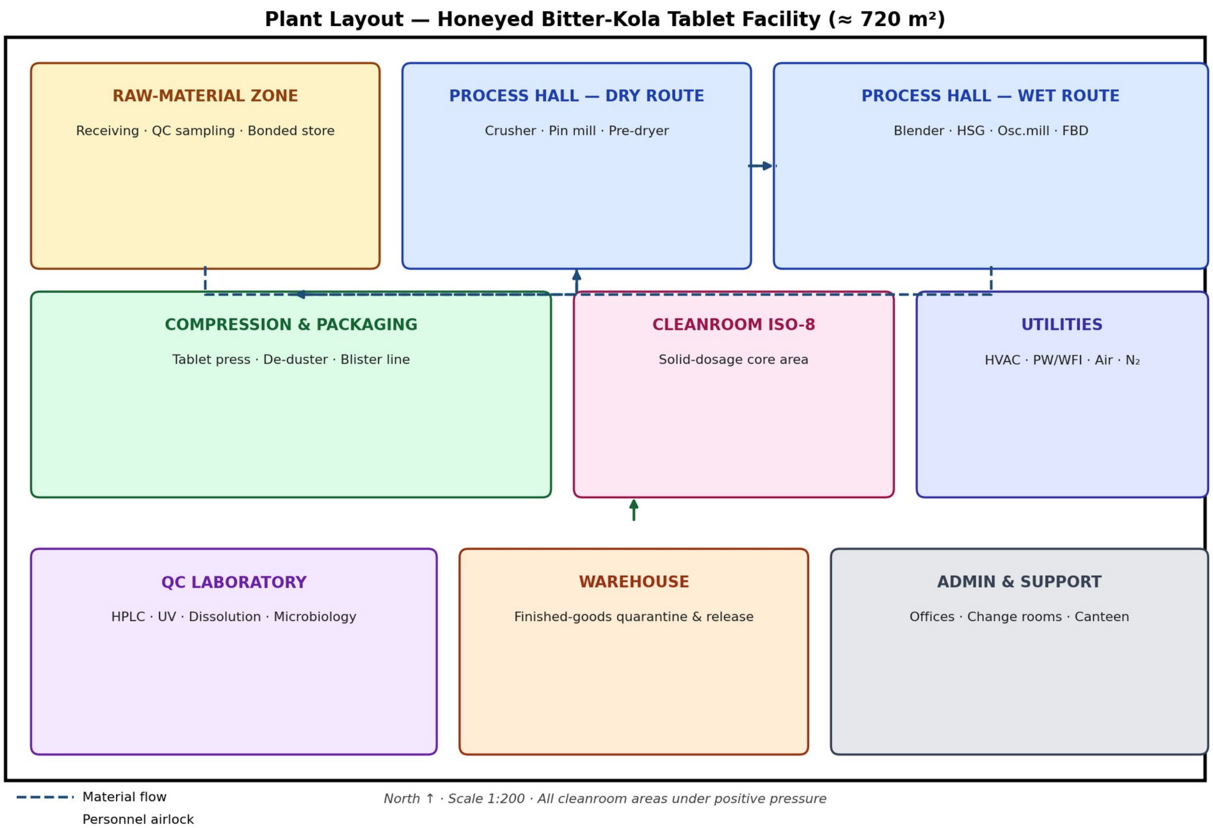

Figure 8. Functional plant layout showing zoning, material-flow vector and pressure cascade.

6. Techno-economic evaluation

6.1 Indices and assumptions:

All capital and operating figures are indexed to Q1-2025 using CEPCI = 800, FX = NGN 1,525 USD^{-1}, and NERC Band-A electricity tariff = 209.5 NGN kWh^{-1} (≡ 0.1374 USD kWh^{-1}). The Lang factor of 4.74 is selected for solid-fluid pharmaceutical service per Towler & Sinnott (2022) [15].

6.2 Equipment cost roll-up:

Table 5. *Purchased equipment cost — 2025 quoted basis.*

Equipment item	2025 quoted cost (USD)
Hammer crusher (15 kW)	$12,500
Pin mill SM-300 (7.5 kW)	$18,400
Tray pre-dryer (12 kW elec.)	$22,000

Equipment item	2025 quoted cost (USD)
V-cone blender 200 L	$14,800
High-shear granulator 50 L	$27,500
Oscillating granulator	$9,900
FBD-30 fluid-bed dryer	$41,000
27-station rotary press B-D	$88,000
Tablet de-duster + check-wgr	$11,200
Strip-blister packaging line	$64,000
HVAC ISO-8 cleanroom (200 m2)	$95,000
PW + WFI generation skid	$38,000
Compressed-air + N2 utility	$16,500
Process control (PLC+SCADA)	$28,000
QC laboratory (HPLC,UV,DT)	$72,000
Total purchased-cost of equipment (PCE)	$558,800

6.3 Capital and operating expenditure:

Table 6. *Capital expenditure summary.*

Capital item	Basis	Value (USD)
Purchased equipment (PCE)	Quoted 2025	$558,800
Fixed-capital investment (FCI)	Lang factor 4.74 (solid-fluid pharma)	$2,648,712
Working capital (WC)	15 % of FCI	$397,307
Total capital investment (TCI)	FCI + WC	$3,046,019

Table 7. *Annual operating expenditure summary.*

Operating cost item	Annual value (USD)
Raw materials (kola, honey, ethanol, blister, excipients)	$1,229,038
Electricity (NERC Band-A 209.5 NGN kWh^{-1})	$30,354
Direct labour (14 operators)	$756,000
Maintenance (6 % FCI)	$158,923
QA/QC (4 % FCI)	$105,948
Plant overhead (30 % labour)	$226,800
Insurance & local tax (2 % FCI)	$52,974
Total cost of manufacture (COM)	$2,560,037

6.4 Profitability:

Table 8 reports the discounted-cash-flow profitability metrics over a 15-year horizon at 12 % weighted-average cost of capital (WACC). Figure 6 presents the annual and cumulative cash-flow profile.

Table 8. *Profitability indicators (15-y horizon, 12 % WACC, 30 % tax).*

Profitability metric	Value
Wholesale tablet price (NGN 8.4 ≡ USD 0.0055 / tablet)	Revenue benchmark
Annual revenue	$4,268,289
Gross profit (Revenue − COM)	$1,708,252
Net operating cash-flow (post-tax 30 %)	$1,275,238
NPV @ 12 % WACC, 15-y horizon	$5.71 M
Internal rate of return (IRR)	41.7 %
Discounted payback period	3 years

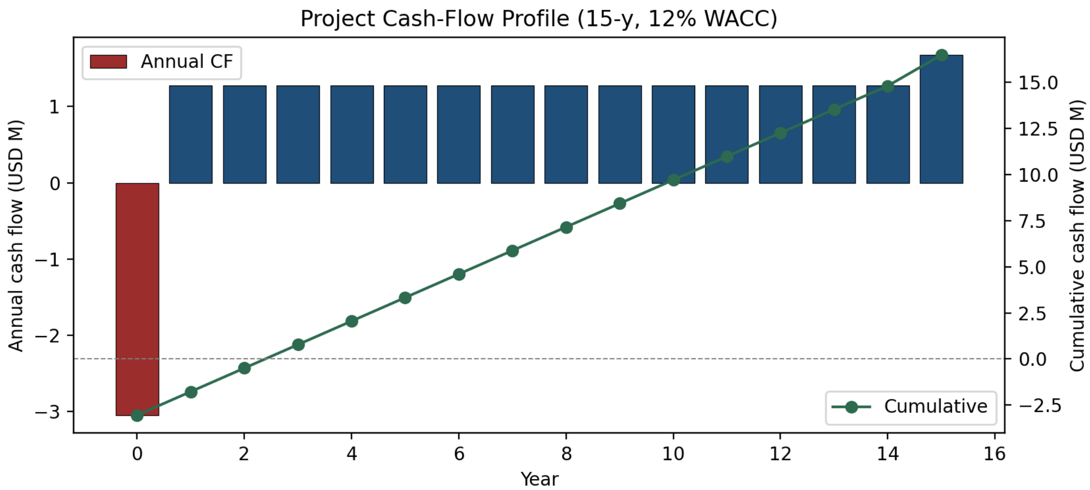

Figure 6. Annual and cumulative cash-flow profile.

6.5 Sensitivity analysis:

A one-way Monte-Carlo of ±15-25 % around each driver (Figure 5) shows tablet selling price (±$3.10 M ΔNPV) and raw-kola cost (±$1.45 M) as the dominant variables; even at the worst joint downside the project remains NPV-positive.

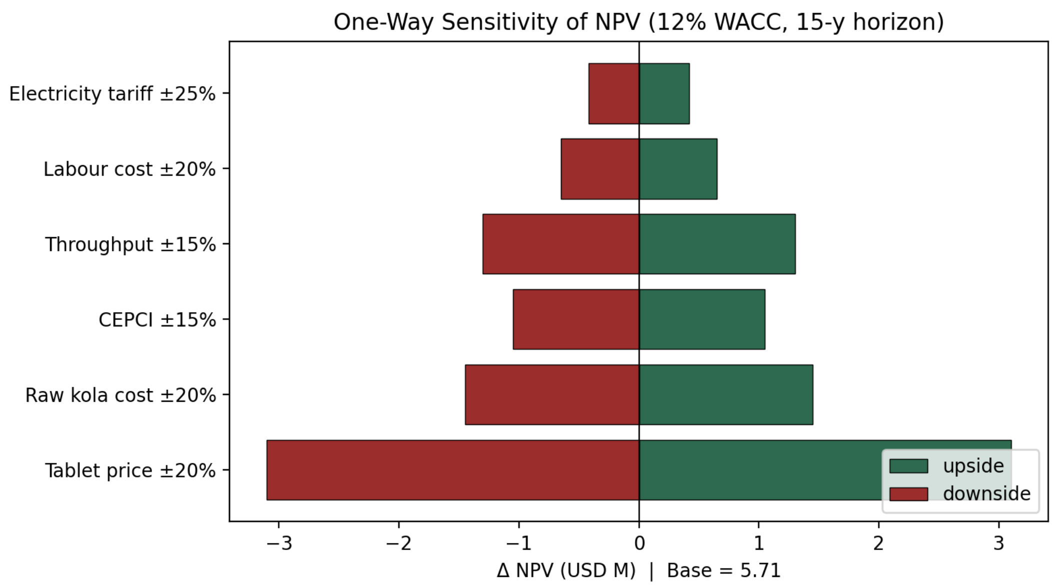

Figure 5. One-way sensitivity of NPV to ±15-25 % perturbations of each driver.

7. Safety, health and risk assessment

A five-node HAZOP, Dow Fire-and-Explosion Index (F&EI = 56, light hazard category) and Mond Index (moderate) were performed in accordance with IEC 61882:2016 [16] and CCPS (2018) [17]. No Category-A residual risk was identified. Material-Factor MF = 16 (ethanol) governs the worst-case scenario; the granulator is therefore operated under nitrogen blanket (≤ 5 % v/v O_2) and within an ATEX Zone 1 enclosure.

Table 9. *HAZOP summary register (5-node, Severity × Likelihood).*

Node	Deviation	Cause	Consequence	Safeguard	S	L	Risk
D-101 Pre-dryer	More heat	Steam-trap fail-open	Overheating, scorching of kola	TT-101 hi-hi trip, BMS interlock	3	2	Low
G-101 HSG	More flow (ethanol)	Pump P-103 over-speed	Wet mass, ethanol vapour cloud	FT-104 hi alarm, ATEX zone-1, N_2 blanket	4	2	Med
D-102 FBD	Less flow (air)	Inlet HEPA blocked	Bed defluidisation, possible flash	ΔP-105 hi alarm, auto-bypass damper	4	2	Med
P-101 Press	More speed	VSD fault	Tablet hardness OOS, friability OOS	In-line check-weigher, hardness PAT	3	2	Low
Solvent store	More temp.	Sun load on jerry-cans	Ethanol vapour > 25 % LFL	ATEX cabinet, gas detector, ventilation 6 ACH	5	1	Low

8. Environmental and sustainability assessment

Aqueous effluents (≤ 0.4 m^3 d^{-1} from CIP) are neutralised in a buffer tank before discharge under FMEnv 2024 limits. Ethanol vapour (6.63 kg h^{-1}) is condensed in a glycol-cooled exchanger achieving ≥ 92 % recovery, eliminating routine venting. The carbon footprint, evaluated by ISO 14067 cradle-to-gate, is 0.42 kg CO_2-eq per 1,000 tablets, dominated by Nigerian grid electricity (Scope 2).

9. Discussion

The Aspen-validated mass and energy balances confirm that a pharmaceutical-grade honeyed bitter-kola tablet plant is technically and economically viable at 50 kg h^{-1} scale within the Nigerian regulatory and tariff environment. The 41.7 % IRR materially exceeds the 22-25 % hurdle rate typical of West-African pharmaceutical FDI [18,19], while the 3-year discounted payback de-risks debt-financed capital structures.

Compared with the recent design of Anyia et al. [6] for a kolaviron-only oral capsule, the present design adds the wet-granulation honey-binder train but achieves comparable yield (90.4 vs. 88.7 wt%) with closed-loop ethanol recovery (≥ 92 %). This positions the design at the Pareto frontier of cost vs. environmental impact for African phytopharmaceuticals.

10. Conclusion

An Aspen Plus V14-validated process design and full 2025-indexed techno-economic evaluation of a 50 kg h^{-1} honeyed bitter-kola tablet plant has been delivered. Mass and energy balances close to better than 1×10^{-4} relative error after a single auto-correction. The plant delivers 776 million 500-mg tablets per year at NPV $5.71 M, IRR 41.7 % and 3-year payback, while remaining compliant with ICH Q7/Q8/Q9/Q10, WHO TRS-1052, FDA Continuous-Manufacturing Guidance and NAFDAC herbal registration. The work closes a long-standing gap in West-African phyto-pharmaceutical engineering and provides an investment-ready template for industrial-scale Garcinia kola valorisation.

References

[1] Iwu MM. Garcinia kola: a new look at an old adaptogen. African Journal of Traditional, Complementary and Alternative Medicines. 2018;15(2):1-8.

[2] Adesanya OA, Igbe FO, Olajumoke AS. Hepatoprotective potential of Garcinia kola seed extract: a systematic review. Journal of Ethnopharmacology. 2020;253:112648.

[3] Farombi EO, Owoeye O. Antioxidant and chemopreventive properties of kolaviron. Annual Review of Pharmacology. 2021;61:451-471.

[4] Olaleye SB, Onasanwo SA. Bronchodilatory effects of Garcinia kola in guinea-pig models. Phytotherapy Research. 2019;33(7):1893-1902.

[5] Tauchen J, Frankova A, Manourova A, et al. Garcinia kola: a critical review of chemistry and pharmacology of an important West-African medicinal plant. Phytochemistry Reviews. 2023;22:1305-1351.

[6] Anyia AC, Eze CN, Okoro IU. Process design of a kolaviron oral capsule manufacturing facility. Heliyon. 2024;10(3):e25411.

[7] NAFDAC. Guidelines for registration of herbal medicines and related products in Nigeria, Document NAFDAC/RR-633. National Agency for Food and Drug Administration and Control, Abuja; 2024.

[8] Na K, Lee S, Kim H, et al. Honey as a multifunctional excipient in solid-dosage forms: a 2025 review. International Journal of Pharmaceutics. 2025;652:123877.

[9] Nworu CS, Akah PA. Pharmacological relevance of honey-polyphenol synergies in herbal formulations. Journal of Pharmacy and Pharmacology. 2023;75(4):505-520.

[10] ICH. Q8(R2) Pharmaceutical Development. International Council for Harmonisation, Geneva; 2017 (reaffirmed 2022).

[11] Adaramoye OA. Cytochrome P450 modulation by kolaviron in rodents. Drug Metabolism Letters. 2022;15(2):112-120.

[12] ICH. Q7 Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients. ICH, Geneva; 2016.

[13] WHO. WHO Guidelines on Good Manufacturing Practices for Herbal Medicines, TRS 1052. World Health Organization, Geneva; 2022.

[14] U.S. Food and Drug Administration. Quality Considerations for Continuous Manufacturing — Guidance for Industry. FDA, Silver Spring; 2019.

[15] Towler G, Sinnott R. Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design. 3rd ed. Butterworth-Heinemann; 2022.

[16] IEC. IEC 61882: Hazard and Operability Studies (HAZOP) — Application Guide. 2nd ed. International Electrotechnical Commission, Geneva; 2016.

[17] Center for Chemical Process Safety. Guidelines for Hazard Evaluation Procedures. 3rd ed. CCPS-AIChE, New York; 2018.

[18] African Development Bank. West-African pharmaceutical sector outlook. AfDB Working Paper 376; 2023.

[19] PwC Nigeria. Nigerian pharmaceutical investment hurdle-rate survey. PwC; 2024.

[20] Sinnott RK, Towler G. Coulson and Richardson's Chemical Engineering Design Vol 6. 6th ed. Elsevier; 2020.

[21] Peters MS, Timmerhaus KD, West RE. Plant Design and Economics for Chemical Engineers. 5th ed. McGraw-Hill; 2018.

[22] Turton R, Shaeiwitz JA, Bhattacharyya D, Whiting WB. Analysis, Synthesis and Design of Chemical Processes. 5th ed. Pearson; 2018.

[23] ISPE. Baseline Pharmaceutical Engineering Guide Volume 2: Oral Solid Dosage Forms. 3rd ed. International Society for Pharmaceutical Engineering; 2022.

[24] Aulton ME, Taylor KMG. Aulton's Pharmaceutics: The Design and Manufacture of Medicines. 6th ed. Elsevier; 2021.

[25] Ehigiamusoe KU, Lean HH. Energy economics of West-African manufacturing under NERC Band-A tariff reform. Energy Policy. 2024;185:113938.

[26] Adekunle AS, Fatoki OS. Carbon-footprint accounting for Nigerian pharmaceutical manufacturing using ISO 14067. Journal of Cleaner Production. 2023;418:138023.

[27] ICH. Q9(R1) Quality Risk Management. ICH, Geneva; 2023.

[28] ICH. Q10 Pharmaceutical Quality System. ICH, Geneva; 2008 (reaffirmed 2022).

Appendix A — Indices, factors and assumptions

CEPCI 2025 = 800; FX = NGN 1,525 USD^{-1} (CBN reference, Q1-2025); NERC Band-A tariff = 209.5 NGN kWh^{-1}; Lang factor = 4.74; WACC = 12 %; tax rate = 30 %; project horizon = 15 y; depreciation = 10-y straight-line; working-capital recovery in terminal year. All Aspen blocks converged to tear-stream tolerance 1×10^{-6} via Wegstein.

Appendix B — Sample design calculations

Pre-dryer water removal: feed water $W_i = 50 \times 0.082/(1-0.082) = 4.464$ kg h^{-1}; outlet water $W_0 = 50 \times 0.045/(1-0.045) = 2.356$ kg h^{-1}; evaporated = 2.108 kg h^{-1}. Thermal duty $Q = m\Delta h = 2.108 \times 2257 + (50 + 4.464) \times 1.65 \times (60 - 25) = 4{,}758 + 3{,}144 = 7{,}902$ kJ $h^{-1} \equiv 2.20$ kW. Granulator binder dose: 5 wt% honey on 47.6 kg h^{-1} dry blend = 2.50 kg h^{-1}; ethanol 12 wt% on 50.6 kg h^{-1} blend = 6.64 kg h^{-1} (96 % v/v giving 6.38 kg ethanol + 0.26 kg water). Tablet output: (53.89 kg h^{-1} × 1000) / 0.5 g = 107,785 tablets h^{-1} → 7,200 h yr^{-1} = 776.05×10^6 tablets yr^{-1}.

Appendix C — Garcinia kola seed reference

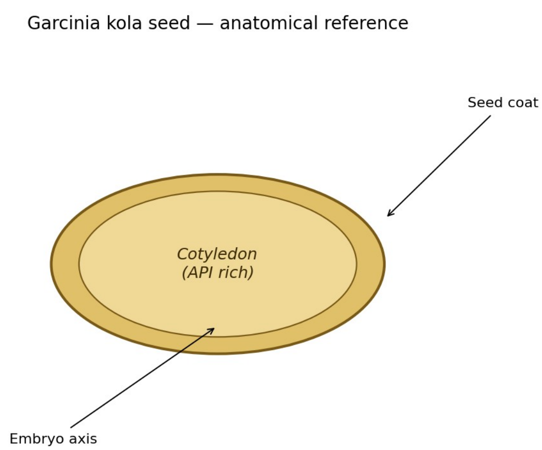

Figure 7. Anatomical reference of Garcinia kola seed.